Databases and Database Users

📌 Key Topics

1️⃣ Basic Definitions 🏗️

- **Database** 🗂️**:** A structured collection of related data, designed for easy access and management.
- **Data** 🔢**:** Raw facts and figures that can be processed into meaningful information.
- **Database Management System (DBMS)** 🖥️**:** Software that helps users store, retrieve, and manipulate data efficiently.
- **Database System** 🌎**:** A combination of the DBMS and the actual stored data, sometimes including applications.

📌 *ببساطة، قاعدة البيانات هي مكان منظم نخزن فيه البيانات، والـ DBMS هو البرنامج اللي بيخلينا نستخدمها بطريقة سهلة ومرتبة.*

2️⃣ Database Properties 🏛️

- A database represents **real-world aspects** (🌍 *Universe of Discourse - UoD*).
- It has a **logical structure** with meaningful relationships (📊 unlike random data).
- **Designed for specific users & applications** (👨‍💻 ensuring relevance & accessibility).

📌 *يعني قاعدة البيانات مش مجرد تجميع بيانات عشوائية، لازم يكون ليها هدف وناس بيستخدموها في شغلهم اليومي.*

3️⃣ Types of Databases and Applications 🔄

Traditional Databases 📜

- Store **numeric and textual** data (e.g., 📊 banking records, employee databases).

Modern Database Applications 🌍

- 🎥 **Multimedia Databases** – Store images, audio, and video.
- 🗺️ **Geographic Information Systems (GIS)** – Handle spatial data for maps & navigation.
- 🧬 **Biological & Genome Databases** – Used in medical & genetic research.
- 🏢 **Data Warehouses** – Store historical business data for decision-making.
- 📱 **Mobile Databases** – Support cloud-based & mobile applications.
- ⏳ **Real-time Databases** – Used in stock trading, IoT, & critical systems.

📌 *قاعدة البيانات مش بس جداول وأرقام، دلوقتي بقت تشمل الصور، الفيديوهات، وحتى البيانات الجينية!* 🤯

4️⃣ Impact of Databases and Database Technology 🌐

📌 **Databases are Everywhere!** 🚀

- 🏦 **Business & Finance:** Banking, insurance, retail, healthcare.
- 💼 **Service Industries:** E-commerce, real estate, law firms.
- 🏫 **Education:** E-learning platforms, student records.
- 📈 **Recent Trends:**
  - Social Media 📱
  - Environmental & Scientific Research 🌱
  - Medicine & Genetics 🧬
  - Smart & Personalized Applications 🤖

📌 *البيانات هي الوقود اللي بيشغل معظم الشركات والتطبيقات الحديثة!* 🔥

5️⃣ Database System Environment and Functionality

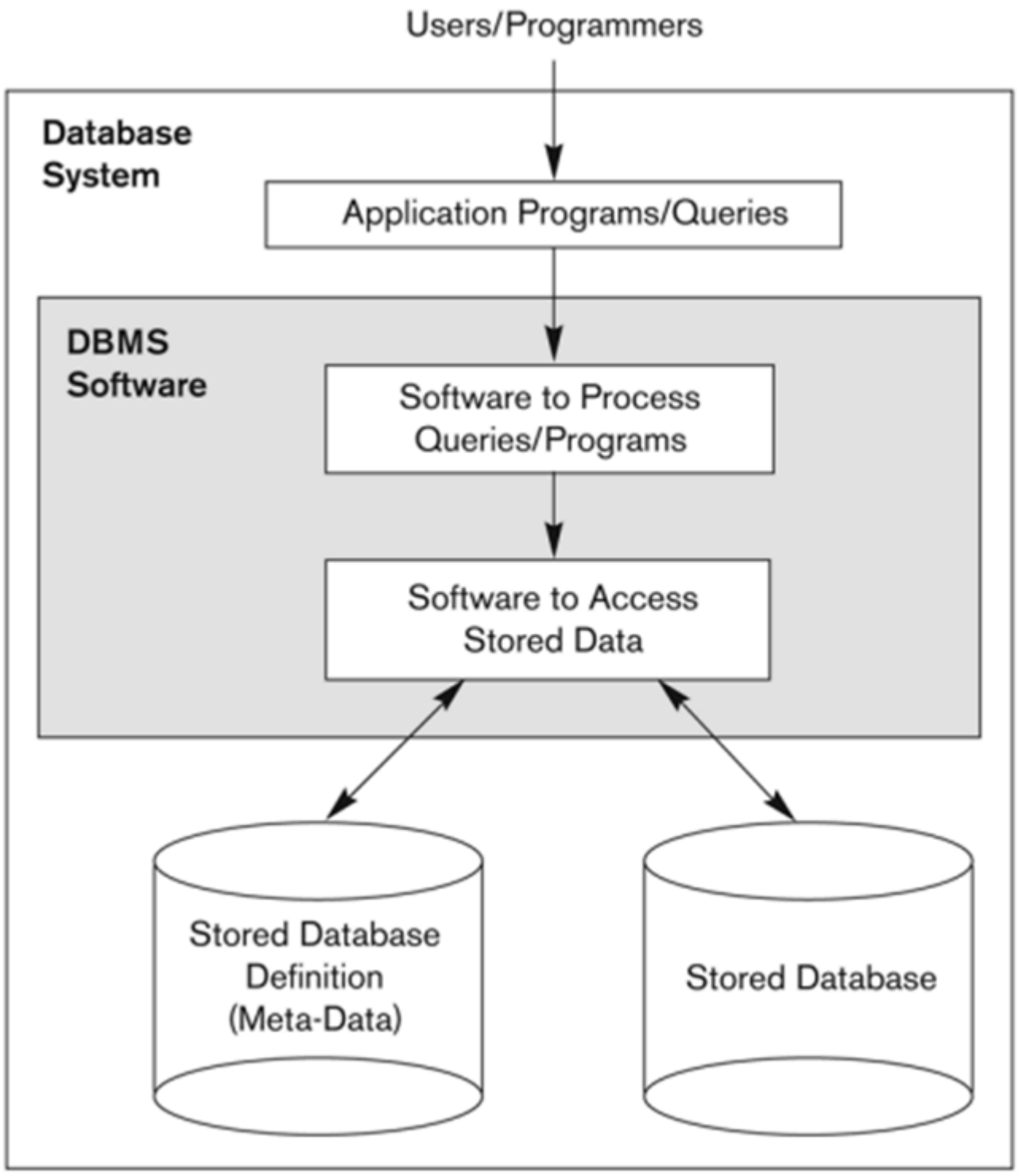


Figure 1.1
A simplified database system environment.

📌 What does a DBMS do? 🛠️

- **Defining** the database (🗂️ data types, relationships, constraints).
- **Constructing** & **loading** initial data 📥.
- **Manipulating data**:
  - 🔎 *Retrieval* – Running queries & generating reports.
  - 🖍️ *Modification* – Inserting, deleting, updating records.
  - 🔒 *Access Control* – Protecting data from unauthorized access.
  - 🤝 *Concurrency* – Managing multiple users at the same time.

📌 *باختصار، DBMS هو المسؤول عن كل شيء في التعامل مع البيانات، من التخزين للحماية والمعالجة الفورية.*

6️⃣ Application Activities Against a Database 🎯

- **Queries** 🔍 – Retrieve data based on specific conditions.
- **Transactions** 🔁 – Perform multiple changes while keeping data consistent.

📌 Key Requirements:

- ✅ Only authorized users can access & modify data.
- 🔁 The system must adapt to changing user needs.

7️⃣ Example of a Simple Database – UNIVERSITY System 🎓

📌 Mini-world Example: University Database 🏫

Entities (الكيانات):

- 👨‍🎓 **STUDENTs** – Enrolled in courses.
- 📚 **COURSEs** – Subjects taught at the university.
- 🏛️ **DEPARTMENTs** – Faculty divisions.
- 👨‍🏫 **INSTRUCTORs** – Professors teaching courses.

Relationships (العلاقات بين البيانات):

- **STUDENTs** enroll in **SECTIONs**.
- **INSTRUCTORs** teach **COURSEs**.
- **COURSEs** belong to **DEPARTMENTs**.
- **STUDENTs** major in **DEPARTMENTs**.

📌 *قاعدة بيانات الجامعة بتنظم كل حاجة، من الطلبة للكورسات وأعضاء هيئة التدريس!*

8️⃣ Key Characteristics of the Database Approach 🧬

- **Metadata** 📑 – Stores information about the structure of the data.
- **Data Models** 🏛️ – Define how data is stored & viewed.
- **Views** 👀 – Each user sees only relevant data.
- **Concurrency Control** 🔄 – Ensures safe multi-user access.
- **OLTP** ⚡ – Handles real-time transactions efficiently.

📌 *دي الخصائص اللي بتخلي قواعد البيانات قوية وسريعة ومتوافقة مع احتياجات المستخدمين.*

9️⃣ Types of Database Users 👥

🎭 Actors on the Scene (Direct Users)

- 🛡️ **Database Administrators (DBAs)** – Manage security & performance.
- 📝 **Database Designers** – Define structure & rules.
- 📊 **End-Users** – Use data for queries & updates.
  - Casual Users 🧐 – Occasional access.
  - Naïve Users 🤖 – Regular use (e.g., bank tellers).
- 🛠️ **System Analysts & Developers** – Design & build applications.

🔧 Workers Behind the Scene (Technical Users)

- **Software Developers** 🧗 – Build the DBMS itself.
- **Tool Developers** 🔍 – Create database utilities.
- **System Operators** ⚙️ – Maintain hardware & software.

🔟 Advantages of Using a DBMS 🌟

✅ **Reduces redundancy** – Avoids duplicate data.
✅ **Enhances security** – Restricts unauthorized access.
✅ **Ensures consistency** – Maintains accurate data.
✅ **Optimizes query performance** – Faster data retrieval.
✅ **Supports multiple users** – Safe concurrent access.

📌 *قاعدة البيانات بتسهل علينا الحياة بتوفير نظام آمن وفعال لإدارة البيانات!* 🔥

1️⃣1️⃣ Database Architectures 🏛️

1️⃣ Centralized DBMS 🏢

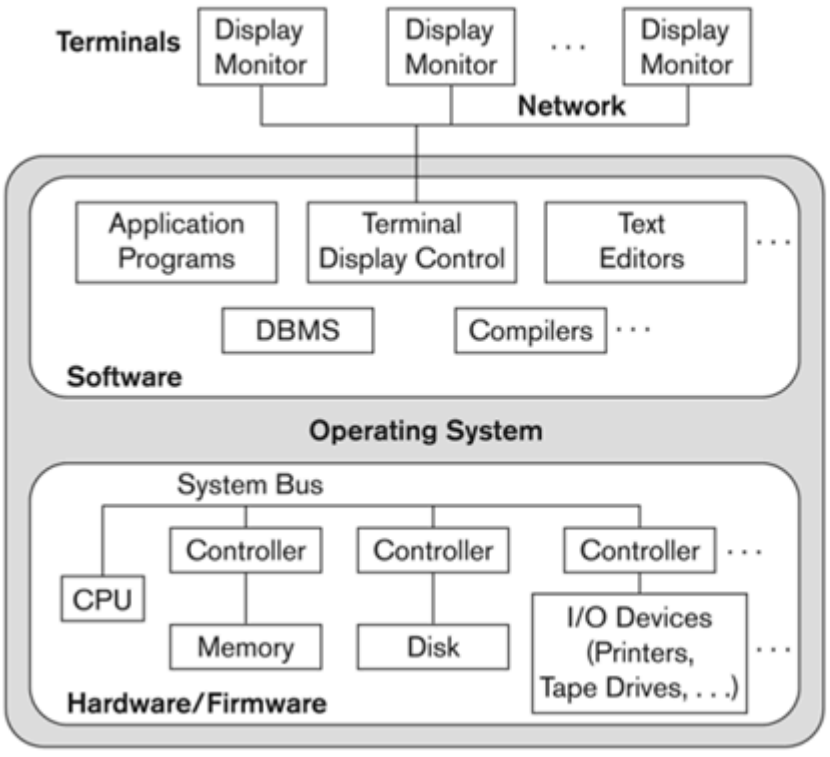


Figure 2.4
A physical centralized architecture.

- 📌 All data & processing happen on a **single** system.
- Users connect remotely (e.g., legacy banking systems).

2 Client-Server DBMS 🖥️📡

- **Clients** send requests.
- **Servers** process queries & store data.
- Used in **SQL databases & cloud systems**.

3 Three-Tier Architecture 🌐

- **Client Layer (UI)** 🎨 – User interacts via a browser.
- **Application Server** 📐 – Business logic & processing.
- **Database Server** 🗄️ – Stores & retrieves data.
- 🔒 **Enhanced Security!** Clients can’t directly access the database.

📌 *دي المعماريات المختلفة لقواعد البيانات، كل واحدة بتناسب احتياجات معينة حسب طبيعة العمل والتطبيقات.*

Figure 2.7
Logical three-tier client/server architecture, with a couple of commonly used nomenclatures.

